

Customer Behavior Analysis Report

Introduction

This report analyzes customer transaction data to understand purchasing behavior, spending patterns, and customer segmentation. The objective is to extract meaningful insights that can help Alfido Tech improve customer engagement and business performance.

The dataset includes information such as total bill amount, tips, customer type, and visit details. Various statistical techniques and visualizations are used to identify trends and patterns in customer behavior.

Dataset Overview

The dataset used for this analysis contains information about customer transactions, including variables such as total bill amount, tip, gender, smoking preference, day, time, and party size.

The dataset consists of approximately 244 records and 7 columns. The data is clean with no missing values, making it suitable for analysis and visualization.

The data types include both numerical (total_bill, tip, size) and categorical variables (sex, smoker, day, time).

Summary Statistics

The dataset was analyzed using basic statistical measures including mean, median, and mode for key numerical variables such as total bill, tip, and party size.

The mean total bill indicates the average spending per transaction, while the median provides the central tendency, reducing the effect of outliers. The mode represents the most frequently occurring values in the dataset.

The analysis shows that customer spending is generally consistent, with moderate variation across transactions. The tip values follow a similar pattern, indicating proportional tipping behavior relative to the total bill.

The party size variable indicates that most transactions involve small groups, suggesting that customers typically visit in limited numbers.

Mean for total bill	19.78594
Mean for tip	2.998279
Mean for size	2.569672
Median for total bill	17.795

Median for tip	2.9
Median for size	2
Mode for total bill	13.42
Mode for tip	2
Mode for size	2
Missing values for total bill	0
Missing values for tip	0
Missing values for size	0

The screenshot shows an Excel spreadsheet with the following data:

	A	B	C	D	E	F	G	H	I	J	K
1	total_bill	tip	sex	smoker	day	time	size				
2	16.99	1.01	Female	No	Sun	Dinner	2		mean for total bill	19.7859	
3	10.34	1.66	Male	No	Sun	Dinner	3		mean for tip	2.99828	
4	21.01	3.5	Male	No	Sun	Dinner	3		mean for size	2.56967	
5	23.68	3.31	Male	No	Sun	Dinner	2		median for total bill	17.795	
6	24.59	3.61	Female	No	Sun	Dinner	4		median for tip	2.9	
7	25.29	4.71	Male	No	Sun	Dinner	4		median for size	2	
8	8.77	2	Male	No	Sun	Dinner	2		mode for total bill	13.42	
9	26.88	3.12	Male	No	Sun	Dinner	4		mode for tip	2	
10	15.04	1.96	Male	No	Sun	Dinner	2		mode for size	2	
11	14.78	3.23	Male	No	Sun	Dinner	2		missing values for total bill	0	
12	10.27	1.71	Male	No	Sun	Dinner	2		missing values for tip	0	
13	35.26	5	Female	No	Sun	Dinner	4		missing values for size	0	
14	15.42	1.57	Male	No	Sun	Dinner	2				
15	18.43	3	Male	No	Sun	Dinner	4				
16	14.83	3.02	Female	No	Sun	Dinner	2				

The dataset was analyzed using statistical measures such as mean, median, and mode for key numerical variables including total bill, tip, and party size.

The mean total bill is **19.79**, indicating the average spending per transaction, while the median value of **17.80** suggests that most transactions are slightly below the average. The mode value of **13.42** shows the most frequently occurring bill amount.

The average tip is **3.00**, with a median of **2.90** and a mode of **2**, indicating that tipping behavior is fairly consistent among customers.

The average party size is approximately **2.57**, with both median and mode equal to **2**, indicating that most customers visit in small groups.

Missing Values Analysis

The dataset was checked for missing values across all key numerical columns, including total bill, tip, and party size.

The analysis shows that there are **no missing values** in the dataset. This indicates that the data is clean and complete, allowing for accurate analysis and reliable results.

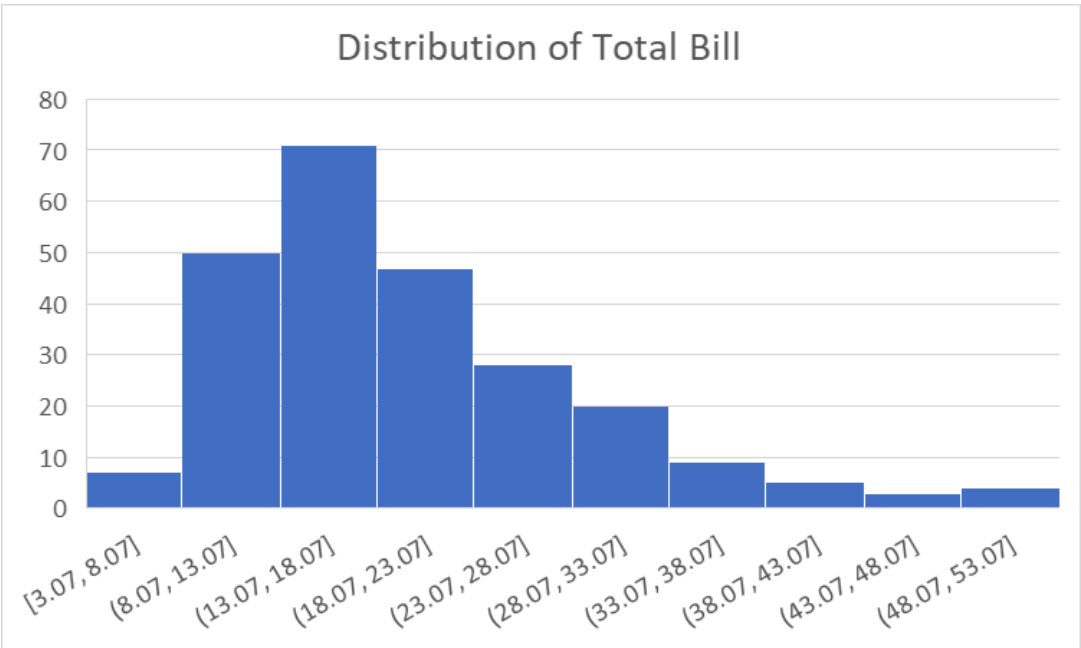
Data Visualizations

Various visualizations were created to better understand customer behavior and spending patterns.

A **histogram of total bill** was used to analyze the distribution of customer spending. It shows that most transactions fall within a moderate spending range, with fewer high-value transactions.

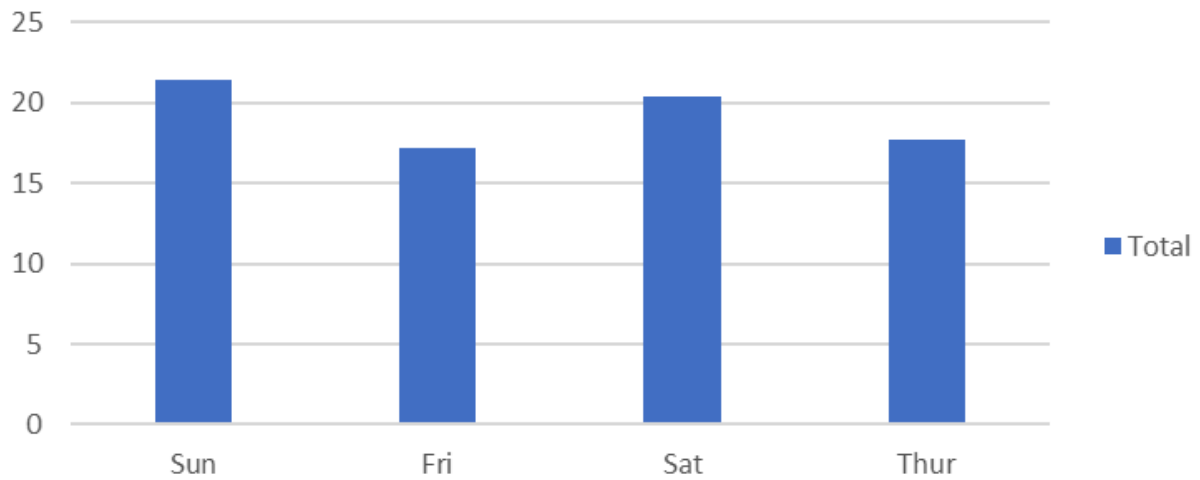
A **bar chart representing average spending by day** highlights how customer spending varies across different days. This helps identify peak days where customer expenditure is higher.

A **line chart showing spending trends by day** provides a visual representation of how spending changes over time, helping to identify patterns and fluctuations in customer behavior.



Average of total_bill

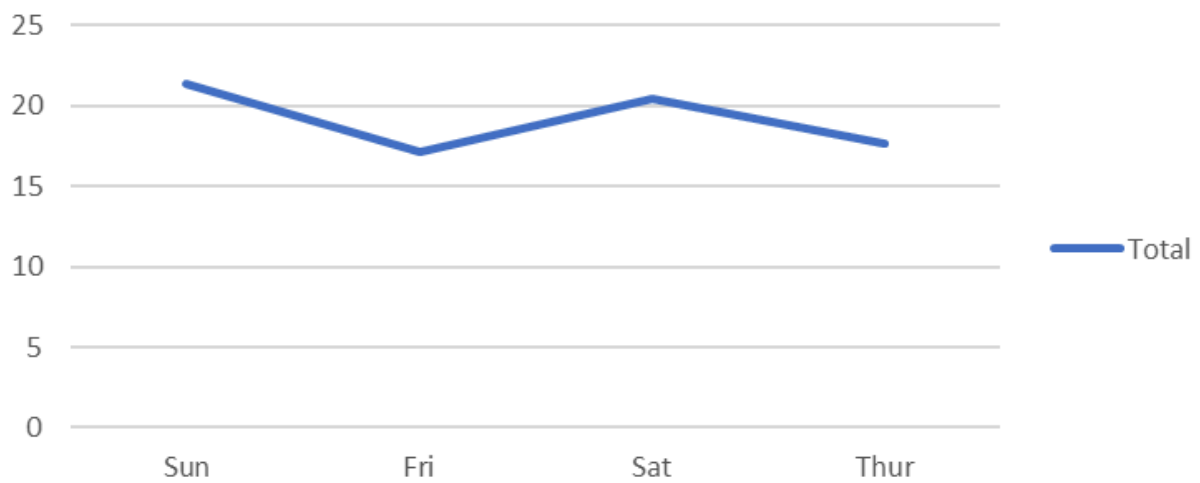
Average Spending by Day



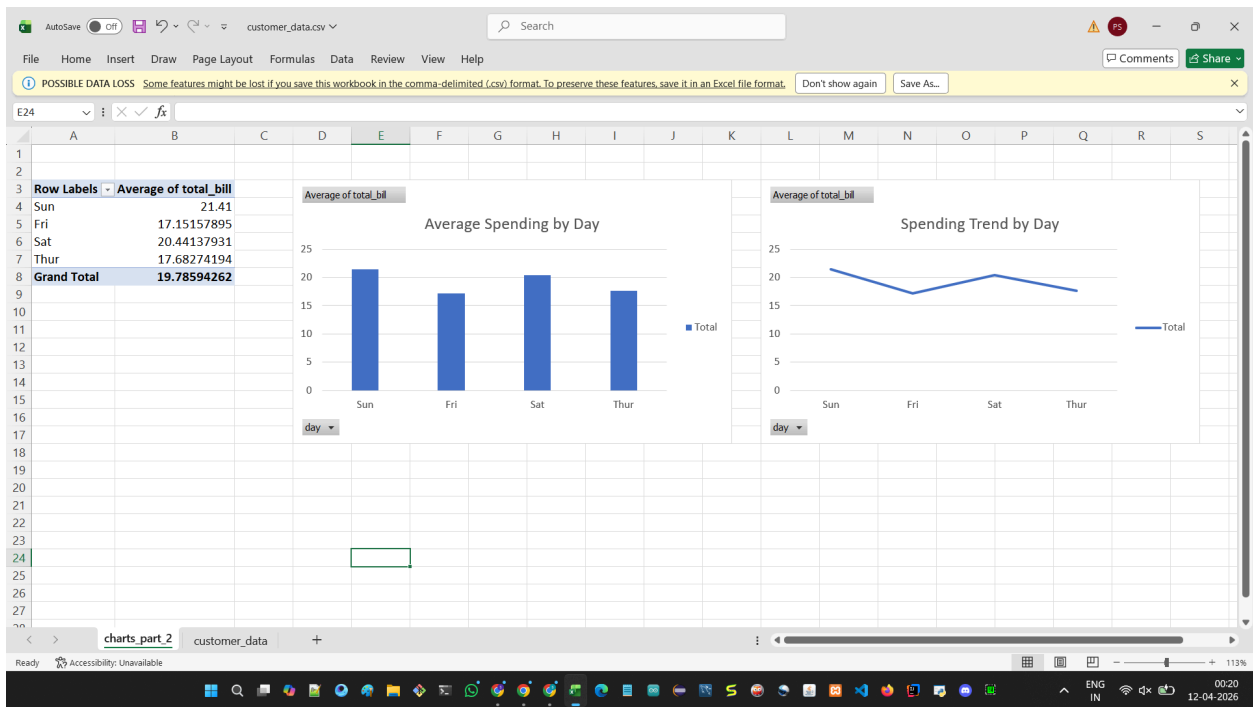
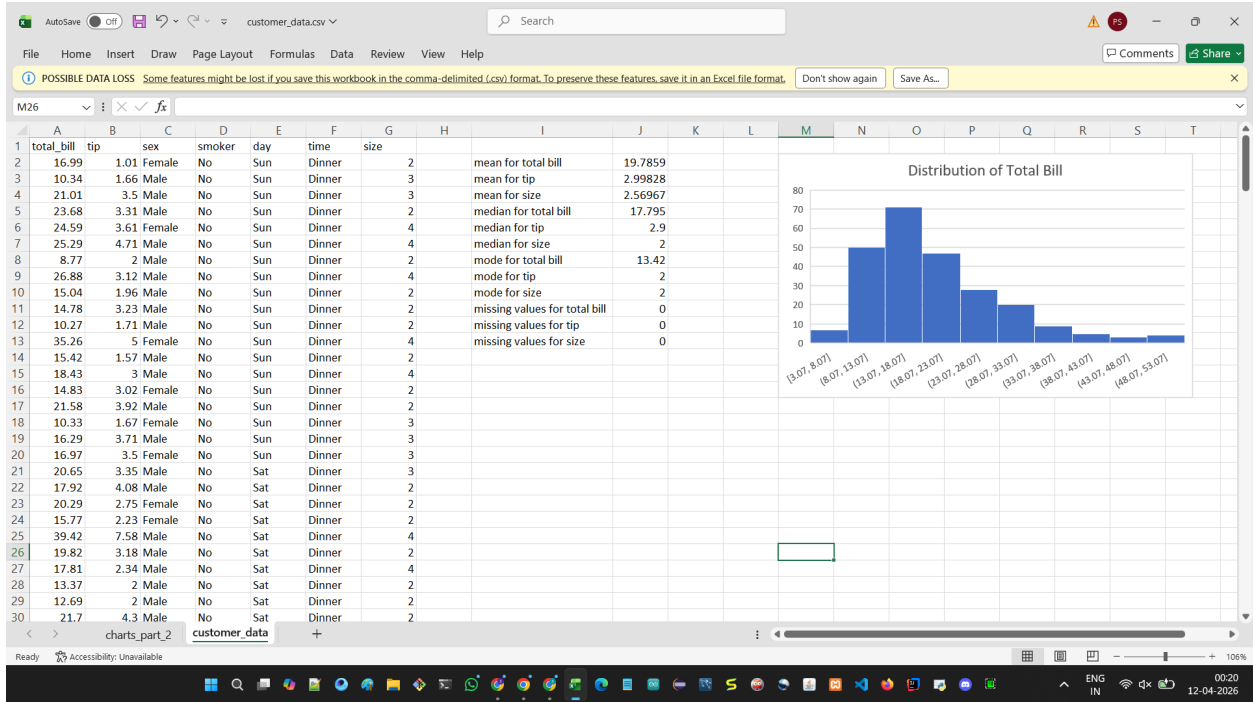
day ▼

Average of total_bill

Spending Trend by Day



day ▼



Customer Insights and Segmentation

Based on the analysis of customer transactions, several key insights were identified.

Firstly, most customers fall within a moderate spending range, with only a few high-value transactions. This indicates that the business relies heavily on regular customers rather than occasional big spenders.

Secondly, customer visits are mostly in small groups, typically of size two, suggesting that the business caters more to individuals or small gatherings.

Thirdly, spending patterns vary across different days, highlighting the presence of peak days where customer expenditure is higher.

Based on spending behavior, customers can be segmented into three categories:

- **High-value customers:** Customers with higher total bill amounts
- **Medium-value customers:** Customers with moderate spending
- **Low-value customers:** Customers with lower spending

This segmentation helps in understanding customer behavior and designing targeted strategies for each group.

Recommendations

Based on the analysis of customer behavior and spending patterns, the following recommendations are proposed for Alfido Tech:

1. **Focus on High-Value Customers**
Offer loyalty programs, exclusive discounts, and premium services to retain customers who spend more.
2. **Encourage Low-Spending Customers**
Provide attractive offers, combo deals, and discounts to increase their spending levels.
3. **Leverage Peak Days**
Increase marketing efforts and promotional campaigns on days with higher customer spending to maximize revenue.
4. **Personalized Marketing Strategies**
Use customer data to provide personalized recommendations and targeted advertisements.
5. **Enhance Customer Engagement**
Introduce feedback systems, email campaigns, and special offers to maintain customer interest and improve retention.